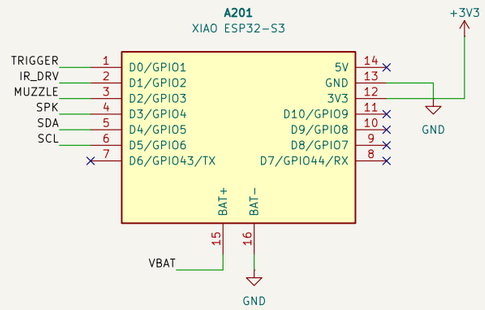
CamoBots — SEEKER BLASTER

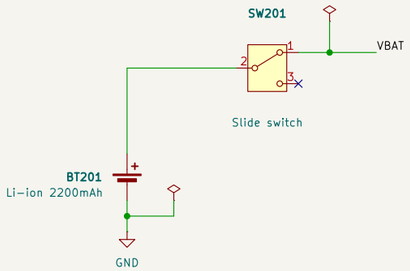
Trigger fires a coded 38 kHz IR shot > muzzle LED flashes > speaker plays the game sounds > OLED shows the HUD

HOST (XIAO ESP32-S3)



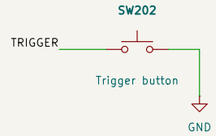
The blaster is also the game host:
it runs the CamoBots hotspot and the web page.
BOOT button doubles as a spare trigger.

BATTERY + SWITCH



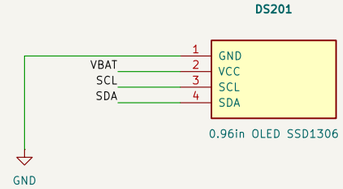
Switch sits in the battery + lead.
VBAT = battery after the switch,
to the BAT+ pad under the XIAO.
Charge through the XIAO USB-C port.
The 5V pin is dead on battery, so
loads that need more than 3.3 V use VBAT.

TRIGGER



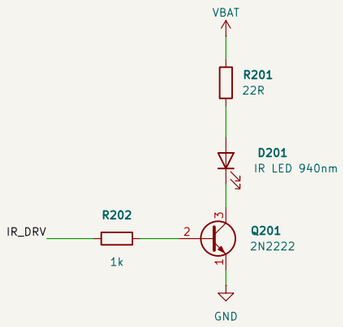
Button between D0 and GND.
No resistor: the XIAO's internal
pull-up holds D0 HIGH.
Press = LOW = fire.
Hold 3 s to stop a round.

OLED DISPLAY (OPTIONAL)



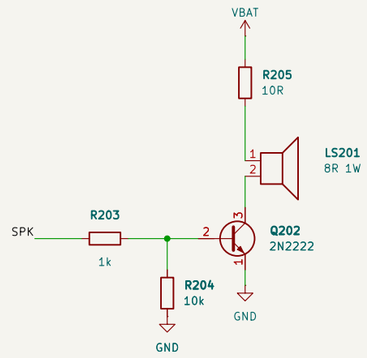
I2C address 0x3C, 100 kHz.
VCC on VBAT: it stays blank on 3V3.
The module has its own I2C pull-ups.
Check the pin order printed on yours:
some boards swap VCC and GND.
No OLED? The game still runs.

IR LED DRIVER (THE SHOT)



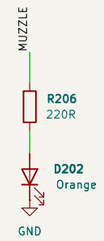
D1 sends the 38 kHz coded burst into the base.
22R sets the LED pulse current (10–15R = longer range).
Long leg (+) of the IR LED goes toward 22R.
A narrow-beam LED (TSAL6100) makes aiming fair.

SPEAKER DRIVER



D3 HIGH switches the transistor on and the speaker sounds.
1k limits base current; 10k holds it off while the XIAO boots.
10R keeps speaker current safe. Speaker + goes to VBAT, not 5V.
2N2222 / S8050: flat face to you, legs down = E B C (check yours).

MUZZLE LED



Flashes on
every shot.
Long leg (+)
toward 220R.

PARTS (BLASTER)

A201	XIAO ESP32-S3 + antenna
D201	940 nm IR LED (TSAL6100)
D202	5 mm orange LED
Q201 Q202	2N2222 or S8050 NPN
SW202	Push button (trigger)
LS201	Speaker 8 ohm 1 W
DS201	0.96in SSD1306 OLED, I2C
BT201	Li-ion 3.7 V 2200 mAh
SW201	Slide switch
R201	22R
R202 R203	1k
R204	10k
R205	10R
R206	220R

Fit the battery and switch last.

Drawn by Abhinav Krishna — CamoBots
Sheet 2 of 2 | designators 2xx
940 nm IR LED on a 2N2222, trigger, muzzle LED, speaker driver, SSD1306 OLED
Runs the Wi-Fi hotspot, ESP-NOW and the game page at 192.168.4.1
CamoBots | Designed by Abhinav Krishna

Sheet: /
File: blaster.kicad_sch

Title: Seeker Blaster + Game Host

Size: A3	Date: 2026-10-27	**Rev: A**
KiCad E.D.A. 10.0.5		Id: 1/1